

# TriZOD — Final Pipeline & Re-Referenced Dataset

Finalized pipeline ·  $\alpha$ -synuclein case study · chemical-shift perturbations

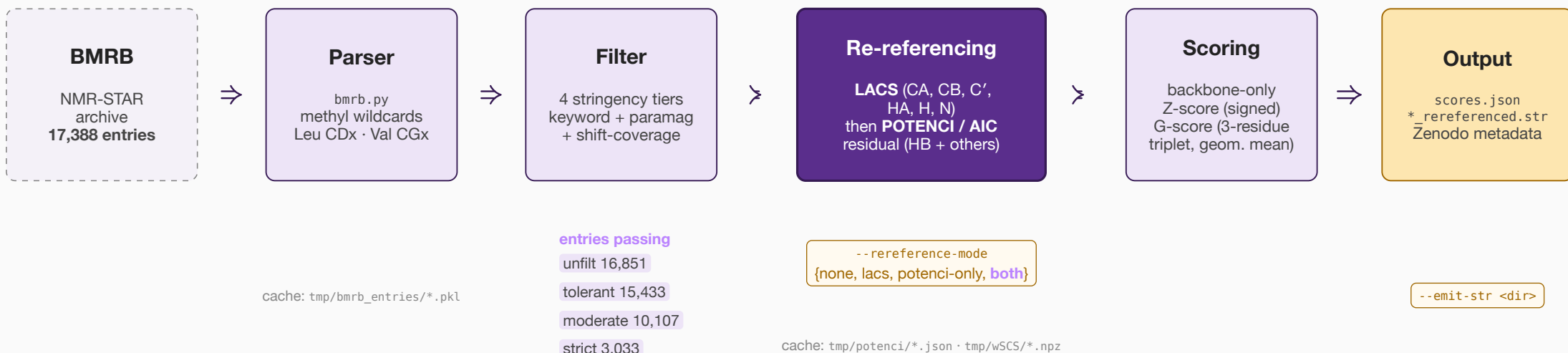
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TUM · TriZOD project meeting

# TriZOD pipeline at a glance

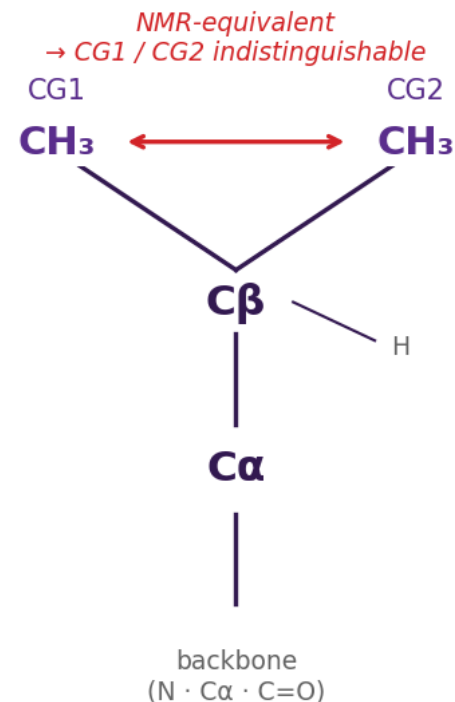


## Next steps:

- exclude multi-molecule (bound) entries that distort G-scores
- per-sequence representative pick (best conditions → median G-score)
- mmseqs2 sequence clustering for ML train/val/test split

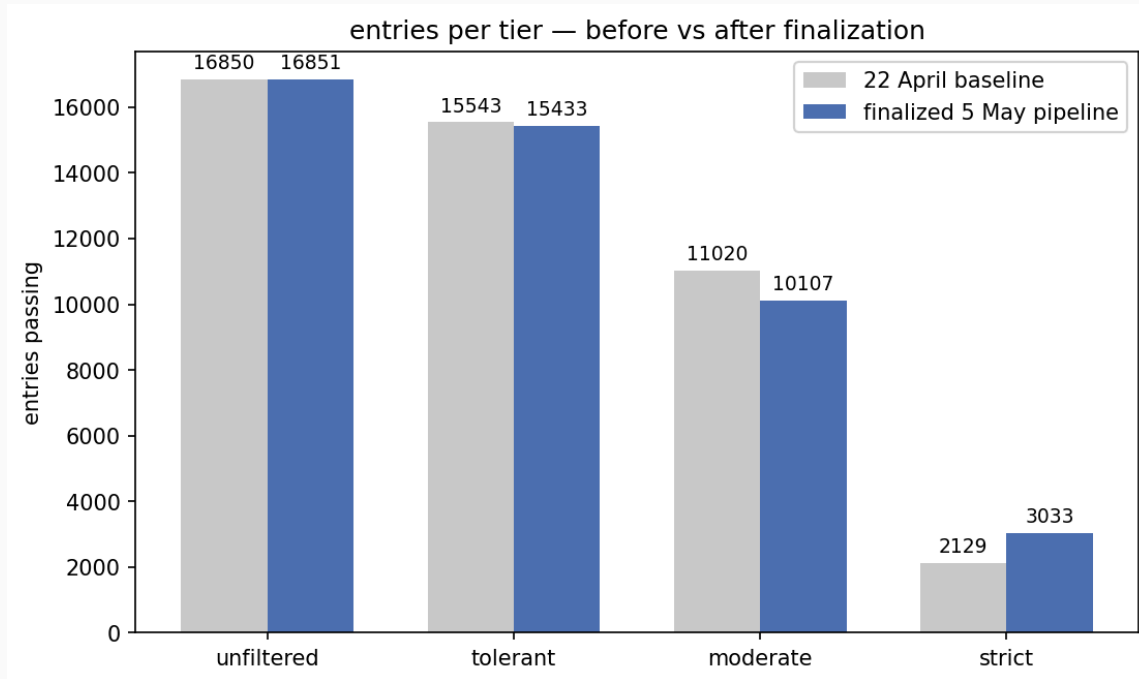
# What's new since 22 April

- Methyl wildcards in the parser (Leu CD1/CD2 → CDx, Val CG1/CG2 → CGx)
- LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default both)
- New flag: `--emit-str <dir>` writes one re-referenced NMR-STAR per scored entity
- **Zenodo deposit metadata** in repo (DOI on first tagged release)



Valine side chain — geminal methyls

# Filter improvements and per-tier dataset deltas



Removed denaturant false-positives, fixed the solid-state regex, added a paramagnetic-sample filter, relaxed `min-backbone-shift-types 5→4` in strict, and broadened the Celsius heuristic.

# Methyl wildcards: what and why

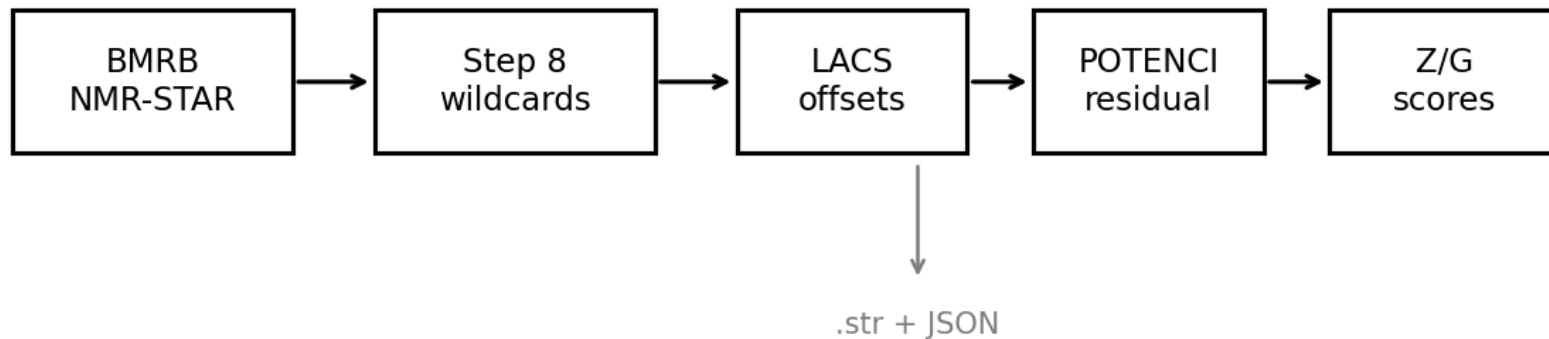
- Leucine: two methyls CD1 + CD2 ( $\gamma$ -carbon).
- Valine: two methyls CG1 + CG2 ( $\beta$ -carbon).
- **Geminal pairs are NMR-equivalent** — most pulse sequences cannot tell them apart.
- BMRB depositors must label one of each pair anyway: arbitrary stereospecificity.

**Before:** LEU CD1 23.4 ppm · LEU CD2 24.1 ppm (false claim of stereospecificity)

**After (ambiguity  $\neq$  1):** LEU CDx 23.4 ppm · LEU CDx 24.1 ppm

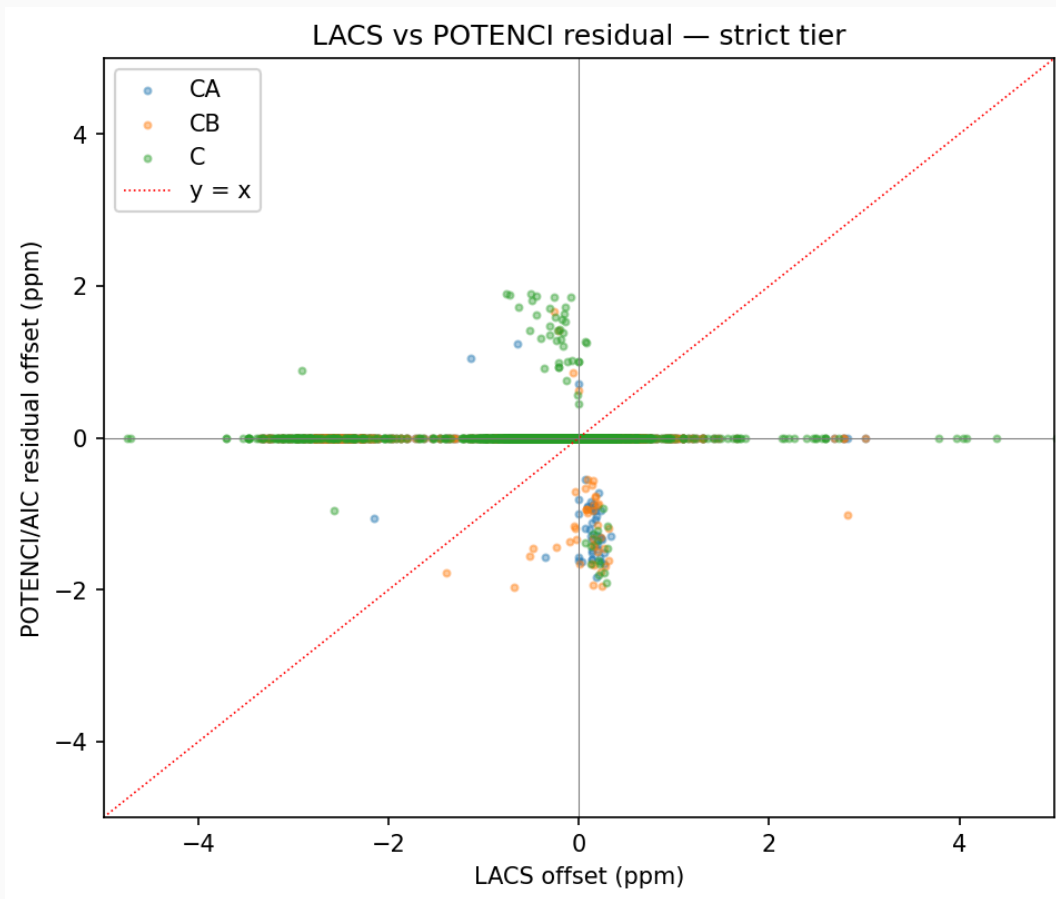
Backbone scoring is unchanged (these are side-chain methyls). The wildcards surface in the emitted `.str` files so downstream auto-assignment tools no longer propagate the false stereospecificity.

# Re-referencing in the pipeline



raw shifts → **LACS pre-correction** (Wishart RC tables, robust line fits) → **POTENCI residual** (AIC-gated rolling 9-window) → Z / G-scores. Default mode is both.

# LACS vs POTENCI residual capture

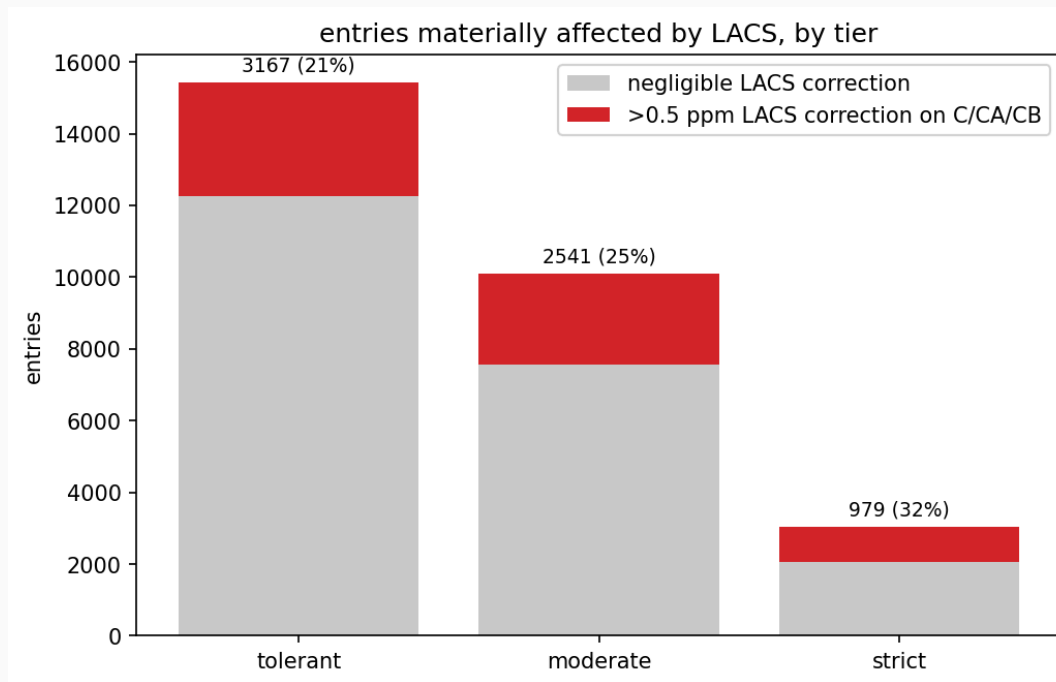


Each dot is **one entry × one atom** (CA / CB / C, strict tier). Not accumulated.

**LACS** catches the large systematic referencing bias.

**POTENCI/AIC residual** mops up the remainder — most points land near the origin once LACS has done its job.

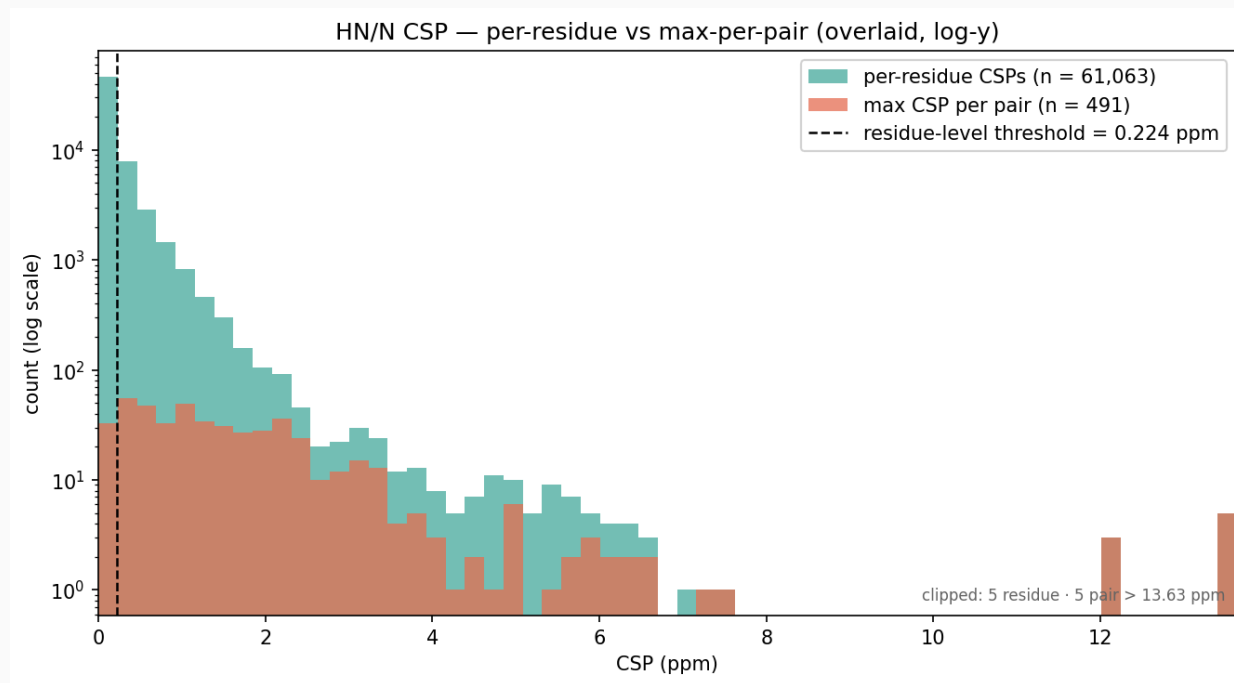
# Entries materially affected by LACS



Red = entries with  $|\text{LACS offset}| > 0.5$  ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for **21%** (tolerant), **25%** (moderate), **32%** (strict).

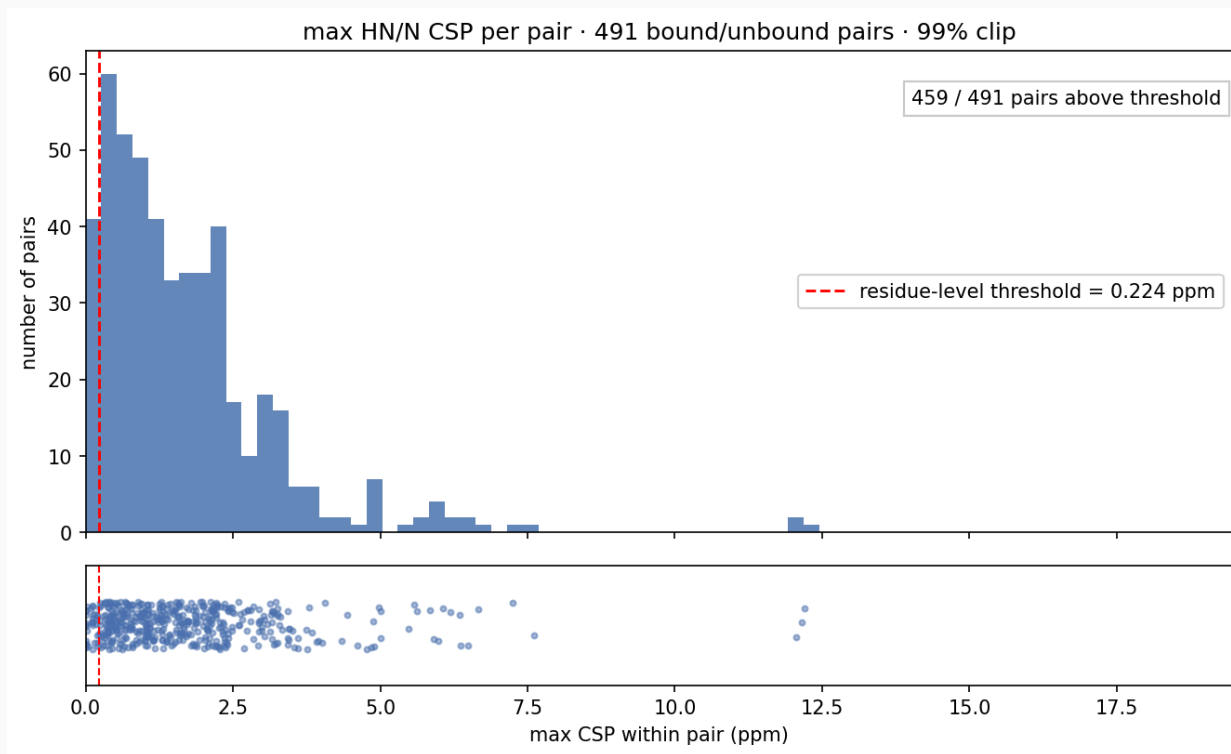


# Chemical-shift perturbation distribution



**Teal:** 61,063 per-residue HN/N CSPs · **orange:** one max CSP per pair (n = 491). Threshold **0.224 ppm** = trimmed-mean+SD on the bottom 90%. Most residues sit below it; almost every pair has at least one residue above. 90 of 581 candidate pairs dropped — no overlapping HN+N coverage between the two entries.

# Max CSP per pair · how strong is the strongest binding shift?



One value per pair (491 points): the **maximum** HN/N CSP within each pair. Pairs below the 0.224-ppm threshold are essentially silent; those above are real binding events.

# What's still TODO

- **Multi-molecule entries distort G-scores** — exclude entries with a non-polymer or nucleic-acid binding partner from the dataset.
- **Per-sequence representative selection** — for the same protein with multiple BMRB entries, pick the one with best experimental conditions; tiebreak by median G-score.
- **mmseqs2 sequence clustering** for the ML train/val/test split (per the original CheZOD/TriZOD report).
- All three are flagged as TODO in the workflow on slide 2.

# Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

## Per-tier counts after the rerun:

| Tier       | Entries | Coverage |
|------------|---------|----------|
| strict     | 3,033   | 17.00%   |
| moderate   | 10,107  | 56.64%   |
| tolerant   | 15,433  | 86.49%   |
| unfiltered | 16,851  | 94.45%   |

# Next steps

- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate on Reid's CSP — extend to multi-atom  $\Delta\omega_{\text{RMS}}$  (Schumann/Williamson 2008)
- Reid follow-up with John Markley on why BMRB stopped LACS reports (frozen July 2020)
- Possible auto-assignment partner who'd benefit from CDx/CGx outputs